Durham University

Examination Paper

Examination Session:	Year:	Exam Code:
May/June	2020	ENGI30620-WE01

Title: MSc Civil Engineering

Environmental Engineering 3

Time Allowed:	2 hours	
Additional Material provided:	None.	
Materials Permitted:	None.	
Calculators Permitted:		Models Permitted: Those from the Casio fx-83 and fx-85 series.
Visiting Students may use dictionaries:	YES	

Instructions to Candidates:	Answer ALL questions. Use a separate answer book for each question. All relevant workings must be shown.

Revision:	

Q1

a) Comment on the following water quality data for four boreholes (see Table Q1) both in terms of how confident you are that the data is correct and how suitable the water would be for drinking.

[50%]

b) Briefly describe where each of these waters might come from or what they have been contaminated with and why you come to this conclusion?

[20%]

c) What important pieces of water quality data are not presented in Table Q1 and why must these parameters be measured in the field?

[30%]

Table Q1. water quality for four boreholes.

	Na (mg/L)	**K (mg/L)**	**Mg (mg/L)**	**Ca (mg/L)**	**Cl (mg/L)**	**SO_4 (mg/L)**	**HCO_3 (mg/L)**	**N-NO_3 (mg/L)**	**BOD_5 (mg/L)**
RAM	Na(23)	K(39)	Mg(24)	Ca(40)	Cl(35)	S(32)O(16)	H(1)C(12)O(16)	N(14)O(16)	
Charge on ion	+1	+1	+2	+2	-1	-2	-1	-1	n/a
Borehole 1	25.3	3.91	2.43	74.1	28	30	256.1	10	2
Borehole 2	248.4	19.6	7.3	8	432.5	24	6.1	1	0
Borehole 3	345	78.2	18.2	80.2	567.2	144.1	183	1	100
Borehole 4	23	82.1	6.1	100.2	28.4	288.2	122	1	0

continued

Q2

a) What does the following simplified Equation (1) represent? What is missing from the equation and without this information, why does the reaction seemingly flout the second law of thermodynamics?

$$\mathbf{CO_{2(g)}} \quad + \quad \mathbf{H_2O_{(Aq)}} \quad \Longrightarrow \quad \mathbf{CH_2O_{(S)}} \quad + \quad \mathbf{O_{2(g)}} \tag{1}$$

[10%]

b) Discuss briefly what is being oxidised and what is being reduced in the reaction (1). What is the relevance of both this reaction and also the reverse reaction to soil health.

[25%]

c) Give a short description of 4 soil functions which are relevant to engineers and explain how these functions relate to soil organic carbon.

[40%]

d) What piece of legislation ensures that engineers have to consider urban soil health? Describe how this current legislation fails to protect the main soil functions in comparison to legislative protection for agricultural soils.

[25%]

continued

Q3

a) The normal depth of flow in a trapezoidal concrete lined channel is $5\ \mathrm{m}$. The channel base width is $7\ \mathrm{m}$ and has side slopes of $1:2$. Manning's coefficient is 0.025 and the bed slope is 0.0001. Determine:

1) The channel discharge flow rate.

[30%]

2) The mean velocity.

[20%]

3) The Reynolds number.

[20%]

b) If the discharge for the channel geometry in **a)** is $30\ \mathrm{m}^3/\mathrm{s}$, find the depth of the flow.

[30%]

END